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Mostafa Hanafy

Senior Full Stack Engineer

Professional Summary

Senior Full Stack Engineer with 10+ years of experience designing, building, deploying, and operating production applications across cloud and on-premise environments. Deep expertise in Node.js, TypeScript, microservices, APIs, PostgreSQL, AWS, Docker, and Kubernetes, with strong production experience in React and Python. Experienced in distributed and event-driven systems, production reliability, CI/CD, and enterprise platforms. Proven technical leadership through architecture decisions, mentoring engineers, and delivering reliable systems across complex multi-environment deployments.

Core Skills & Technologies

Backend: Python, Node.js, TypeScript, FastAPI, NestJS, Express.js, Django, Odoo

Frontend: React, Next.js, React Native, JavaScript (ES6+), HTML, CSS, Redux

Databases: PostgreSQL, MySQL, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Linux, Bash, Nginx

Messaging & Queues: RabbitMQ, AWS SNS, AWS SQS

AI-Assisted Development: GitHub Copilot, Claude Code, OpenAI Codex

Tools & Practices: Git, Jira, RESTful APIs, TDD, Agile, Microservices, Event-Driven Architecture

Experience

Senior Full Stack Engineer | AirCarbon Technology Ltd, UAE | Jan 2022 - Present

- Design and develop scalable Node.js/TypeScript microservices and REST APIs supporting production trading and marketplace workflows.
- Develop and maintain React/TypeScript applications and internal tools, building responsive and accessible interfaces with reusable components, semantic HTML, keyboard navigation, and robust API integration.
- Built Python/FastAPI internal services and automation tools for reporting, operational workflows, and data processing.
- Build event-driven distributed systems using RabbitMQ/SNS/SQS, with idempotency, retry handling, and resilient integration patterns.
- Deploy and operate containerized services across multi-environment cloud infrastructure using Docker, Kubernetes, and CI/CD pipelines.
- Own production troubleshooting and incident response for critical services, using logs, metrics, health checks, and root-cause analysis to improve reliability, performance, and uptime.
- Collaborate with product, QA, and operations from requirements through deployment, while contributing to architecture reviews, code quality, maintainability, and scalability.

Senior System Developer | Dynamic Smart Solutions, UAE | Apr 2018 - Dec 2021

- Developed and customized Odoo ERP solutions using Python, XML, and JavaScript, and built Node.js/React portals and REST APIs to extend ERP capabilities.
- Built and supported bilingual Arabic/English enterprise applications, including RTL interfaces and localized business workflows.
- Designed and implemented attendance and mailing systems integrated with ERP, streamlining internal operations.
- Built a mobile ERP reporting app for real-time analytics.

Freelancer | Upwork, Egypt | Jan 2018 - Apr 2019

- Delivered Odoo and full-stack projects for international clients, including Python-based vending management solutions.

Development Team Leader | Selective Systems, KSA | Jan 2015 - Jan 2018

- Led engineers delivering enterprise Odoo platforms, mentoring developers and managing cloud/on-prem deployments and client integrations.

Teaching Assistant | Information Technology Institute(ITI), Egypt | Oct 2014 - Feb 2016

- Taught Linux, Bash, databases, Python, and open-source development fundamentals.

OpenERP Developer | It-sys Corporation, Egypt | June 2014 - Jan 2015

- Developed and deployed Odoo HR and operations modules and supported end-user training.

Personal Projects

AI Request Layer (Open Source) | GitHub | npm packages available

A modular toolkit for building reliable, observable, and cost-efficient LLM applications in production.

Core Components:

- **prompt-cache** – Caching layer for LLM calls with deterministic keys and in-flight deduplication to reduce cost and latency
- **llm-retry-guard** – Retry wrapper with exponential backoff and jitter for handling transient failures
- **token-budget-guard** – Token usage control to enforce limits and prevent unexpected cost spikes
- **ai-request-logger** – Structured logging for tracking latency, token usage, and estimated cost

ts-package-init (Open Source CLI Tool) | GitHub | npm packages available

TypeScript scaffolding tool supporting backend, library, CLI, monorepo, NestJS, and Moleculer presets with ESM/CommonJS, ESLint, and Prettier.

Education

Open Source Applications Development Diploma | Information Technology Institute (ITI), Cairo, Egypt | 2013–2014

Focused on Python, Django, Linux, Bash, and open-source frameworks.

Bachelor of Engineering (Aeronautics & Aerospace) | Cairo University, Egypt | 2008–2013

Graduation Project: Unmanned Micro Air Vehicle – Grade: Excellent (95%).

Achievements & Links

- Languages: Arabic (Fluent), English (Very Good)